Andrew G Milmoe

Lead UX Product Designer

AI Augmented

Forward Deployed On-Site Research

Enterprise Software in Regulated Environments

Industrial IoT Platforms



From concept to global operations,
I use AI and user-centered design to give the people who run critical systems **superpowers**.

User Research Site Visit



Wind farm maintenance crew interviews in Colorado prairie.

I design decision-making, not just screens

The systems I design exists so someone under pressure can ask a hard question and get a confident answer quickly.

A rocket engineer comparing live telemetry to history.

A care coach documenting a call in real time.

An engineer deciding to take an asset offline.



Drill ship engineer user research off the coast of Louisiana.

Core design principles

“Get out of the building” and observe where the work happens.

Reframe before you solve.

Make the system visible to build trust and alignment.

Design out the dependency.

ML & AI Systems across Industrial IoT, Aerospace, and Healthcare

Aerospace Data Management On-site research with ~30 rocket engineers shifted the roadmap from query optimization to an end-to-end data insights pipeline.

Business Outcomes Research with 35 stakeholders across four countries pivoted GE's Visor strategy, enabling a $180M contract and doubling sales in two years.

UX Design on a Global Scale Guided data visualization strategy for GE Predix APM systems monitoring assets responsible for ~12% of global power generation.

Startup Velocity & Revenue Acceleration Designed AI query tools supporting 135% YoY growth; identified bottlenecks that advanced revenue recognition by ~1 month.



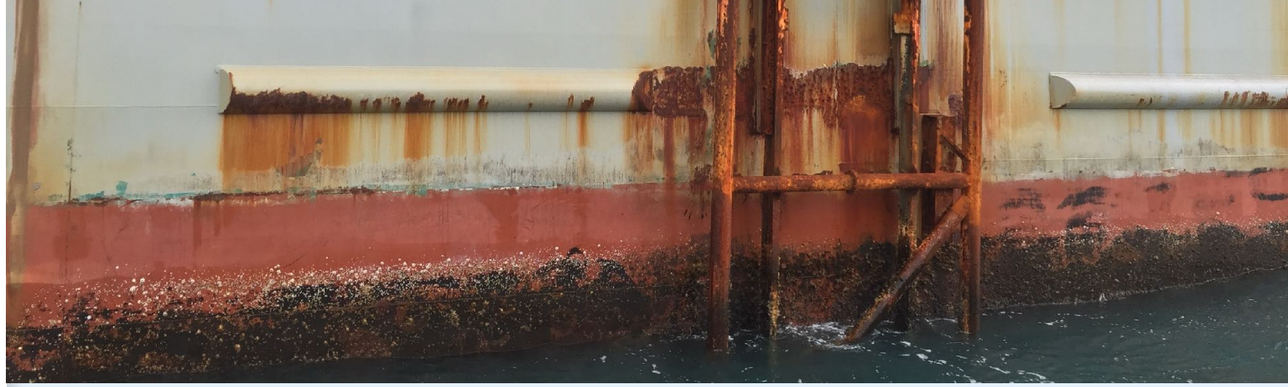




GE DIGITAL – POWER CONVERSION

Industrial IoT Service Portal for Remote Monitoring and Diagnostics

Born as a pilot created by a pair of hardware engineers to enable field service engineers to remotely access marine power systems to reduce warranty claims and avoid $1M per day in unexpected downtime



Challenges

- Hybrid edge/cloud constraint
- Satellite bandwidth limitations
- Need to reduce interruption
- Hardware sales → “as a service”

Key Responsibilities

- On-site User Research
- Rapid Prototyping
- Concept / Vision

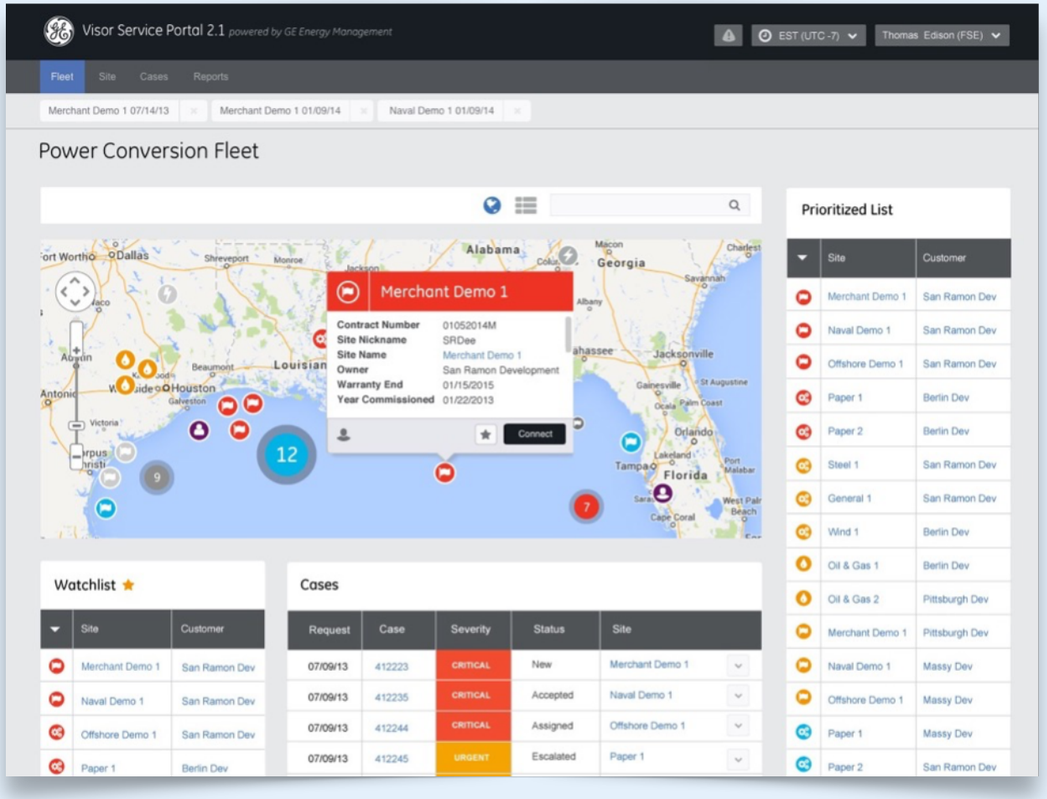
Timeline & Team

90 days to MVP + 90 days to commercial launch.

Product manager, business analyst, two engineers, and myself.

Results

- $180M 10-year contract, doubled sales
- Reduced trip report access from 8 hours to 15 minutes
- Regained the trust of sales team and customers



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CHALLENGE - TIMELINE

My role: lead User Experience Design as the organization shifts from Engineering led to human-centered processes

Engineering Led

Human-Centered

KICK OFF

DAY 90 MVP

DAY 180 LAUNCH

Story Mapping

Prioritized features with the Product Owner and Engineering Team to ensure the greatest impact.

Rapid Prototyping

Initial **concept model** to get the team rolling.

Then **clickable prototype wireframe** iterations for lightweight MVP.

On-Site Research

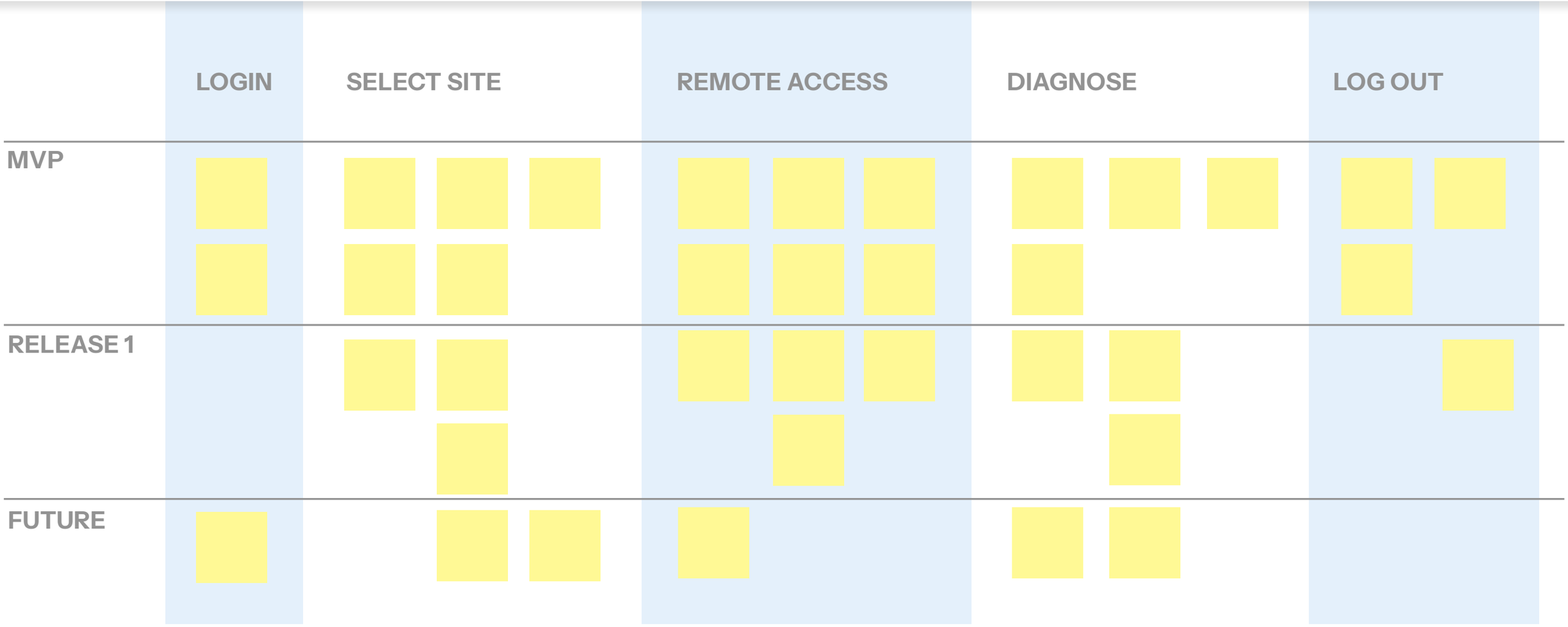
Gained trust with European stakeholders through **participatory design & clickable prototype user testing.**

Vision

Pivoted to a simplified flow. Mocked up **two year vision** to inform system architecture roadmap.

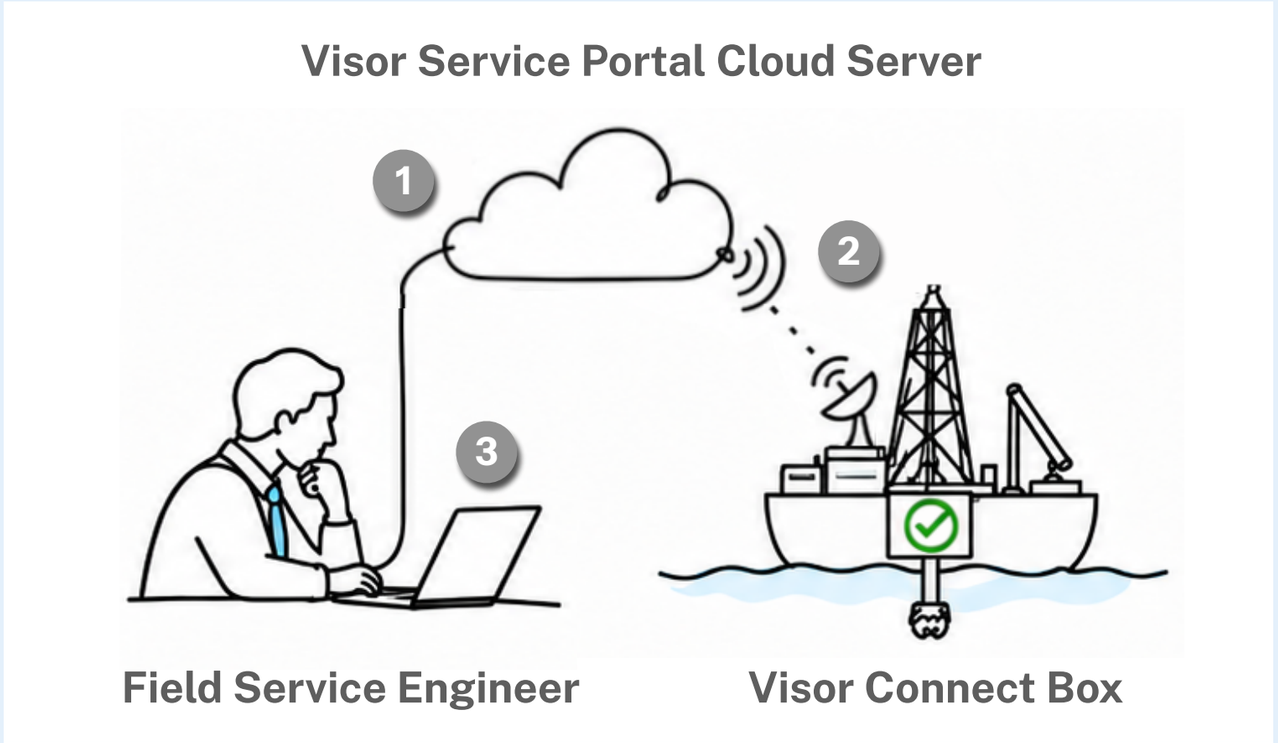
PROCESS - STORY MAPPING

Led story mapping to align Product and Engineering on the critical path to MVP



Created a minimum-feature concept model to test feasibility and guide the commercial architecture

1. **One Field Service Engineer** (FSE) connects to the **Cloud** ...
2. Cloud connects securely to Visor Connect Box aboard **one vessel via satellite phone.**
3. FSE **establishes a remote desktop connection** to monitor the site.



PROCESS - RAPID PROTOTYPING

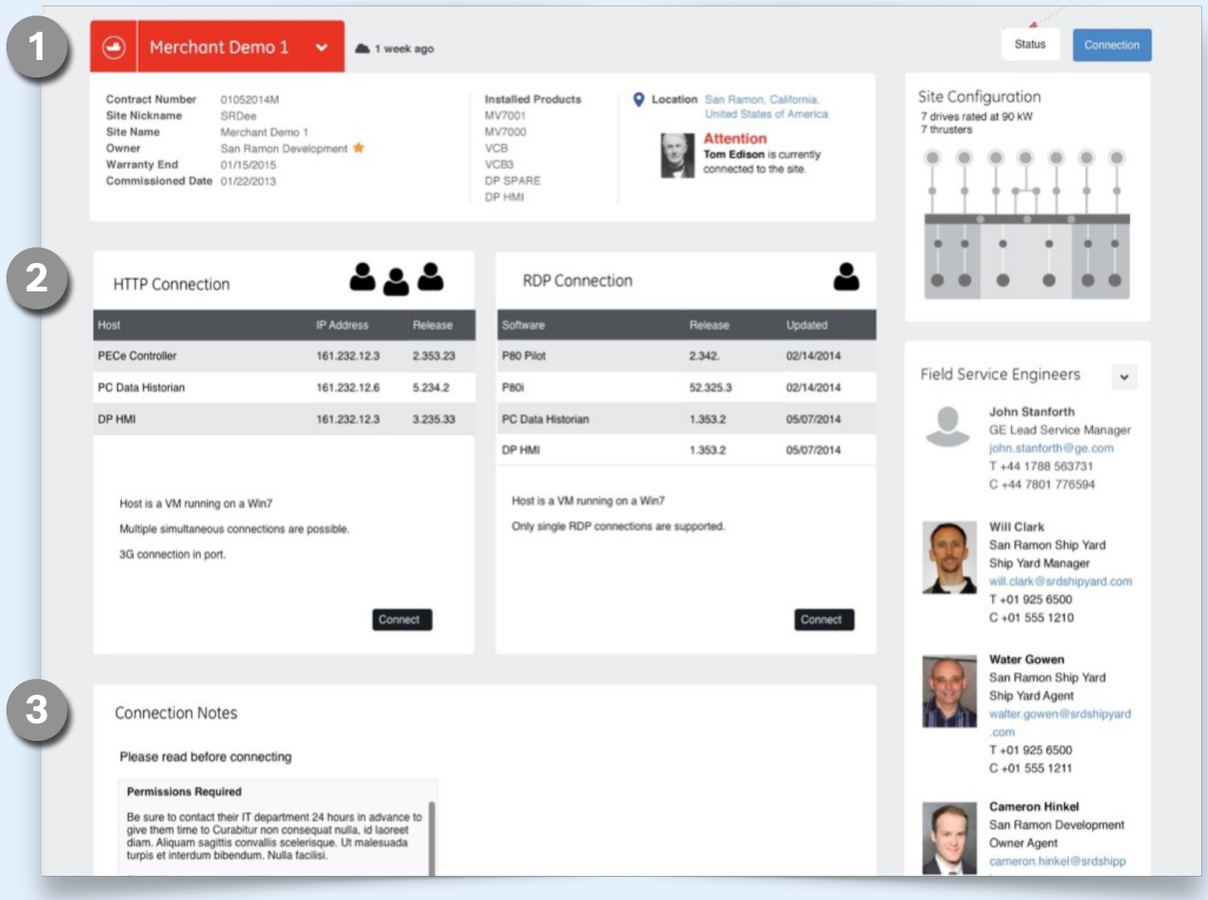
Analyzed Service Portal connection pain points and designed the proof of concept to remove dependencies

1. Clarify asset identity and connection status to prevent user collisions.

2. Surfaced connection metadata to diagnose software incompatibilities.

3. Transcribed institutional knowledge from handwritten notebooks.

Asset Page Proof of Concept



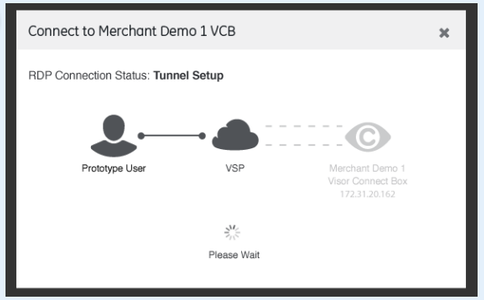
PROCESS - RAPID PROTOTYPING

Designed a visual workflow to make the complex connection process visible and easier to diagnose

1. Confirm cloud connection and internet viability from the road.
2. Verify the secure VPN tunnel. Users have five minutes to connect once secured.
3. Confirm data connection. On success, users switch to diagnostic software.
4. Alert users on disconnect, as leaving the connection open blocks others.

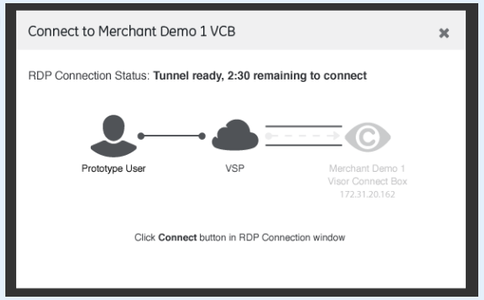
Secure Tunnel Workflow UI

1



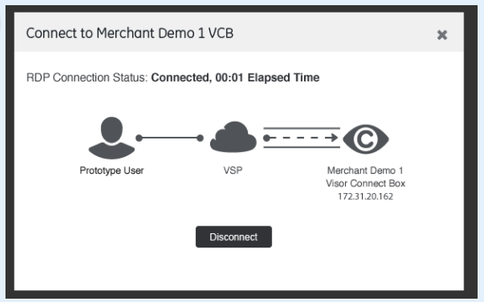
Field Service to Cloud

2



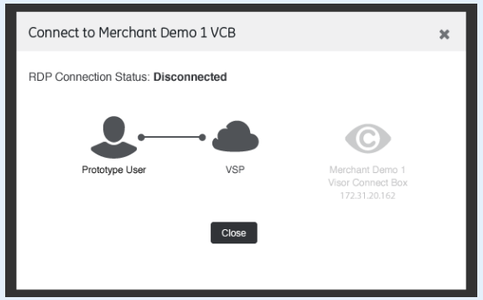
Cloud Tunnel to Ship

3



Data Connection Live

4



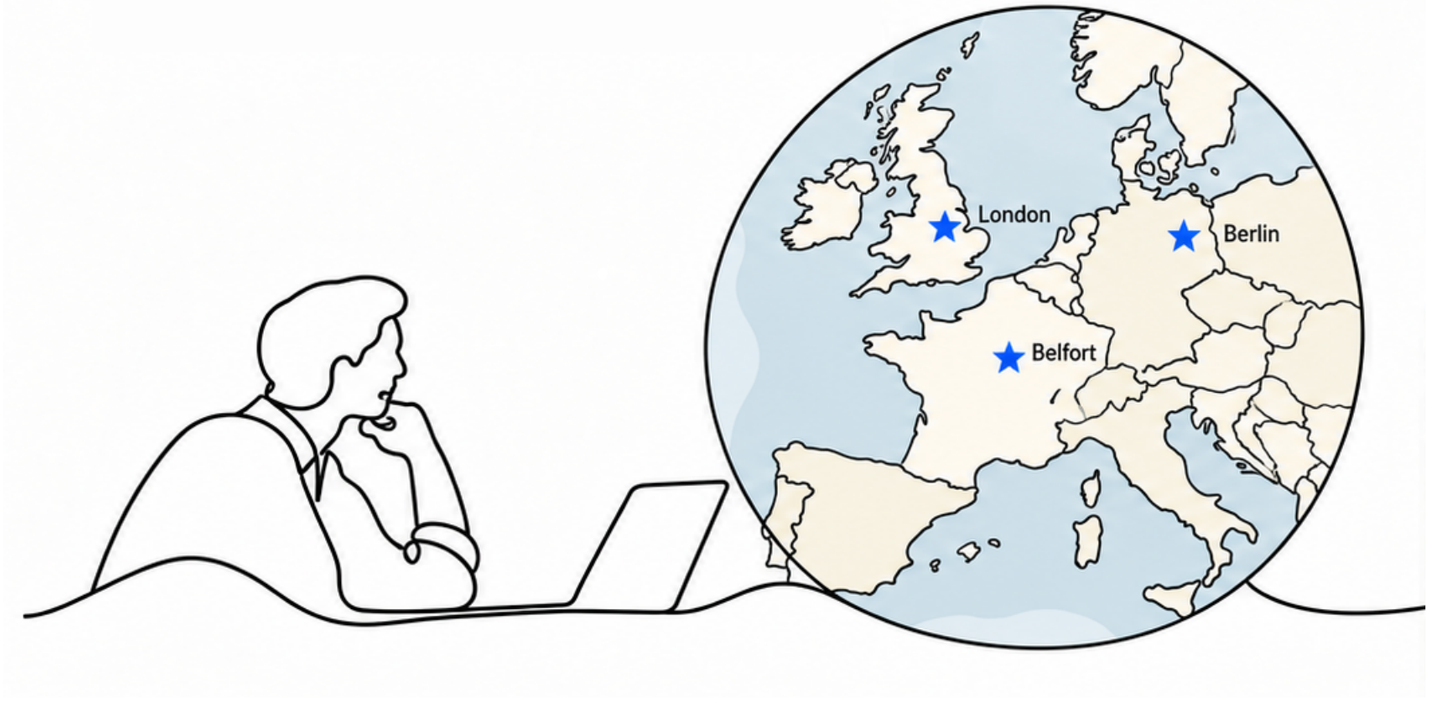
Confirmed Disconnect

PROCESS - ON-SITE RESEARCH

Conducted UX research in the US and 3 countries, 35 interviews with Field Engineers and Managers

Key insights

- Low bandwidth unreliable satellite connectivity
- Remote desktop access creates friction
- Diagnostic data increases speed to action





SOLUTION

Research insights reframed the solution, the pivot from Remote Desktop to trip reports accelerated delivery by 30 days

New Iteration Based on UX Research



Scott receives a failure notice from the VSB, then a link to download the trip report file.

Minutes later he begins diagnosis.



The drill ship engineers can reboot the drive to clear the alarm and **return to their chartered mission.**



Henry is happy the system recovered before he knew it was down.

IMPACT - CUSTOMER FEEDBACK

Customer trust grew once they felt the value of the new Visor system

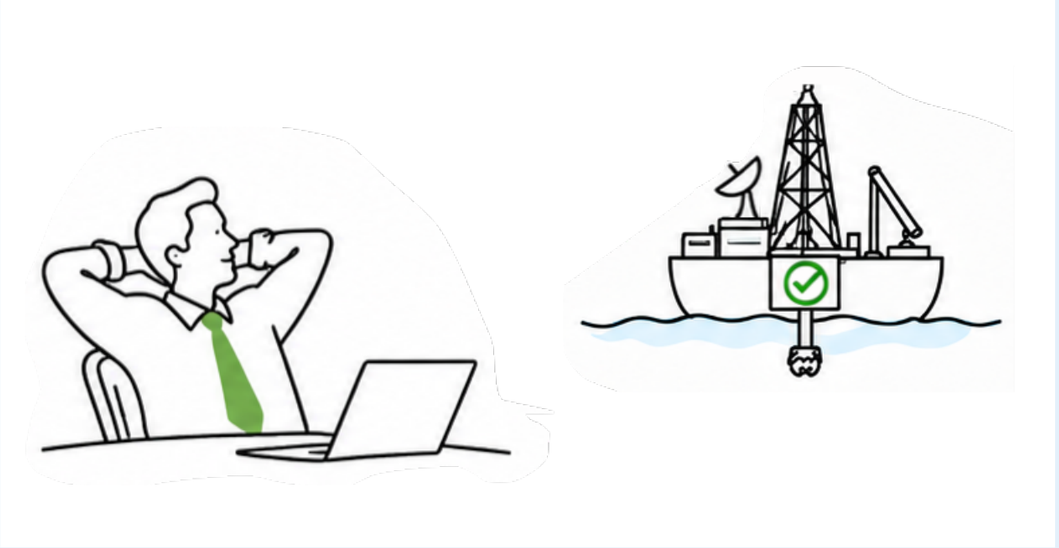
"The Drive Trip Auto Alert feature in Visor is … already proving to be a **very useful tool.**"

The customer's vessel was drifting off course, an alert was received prompting GE to contact the vessel within minutes to offer assistance.

"Access to this data … allowed us to provide advice and guidance to get both drives **back into operation** … so the vessel could complete the transit voyage to Oman."

The customer had parts and engineers ready and waiting.

"The **successful service visit** allowed the vessel to fulfill its charter on time … "



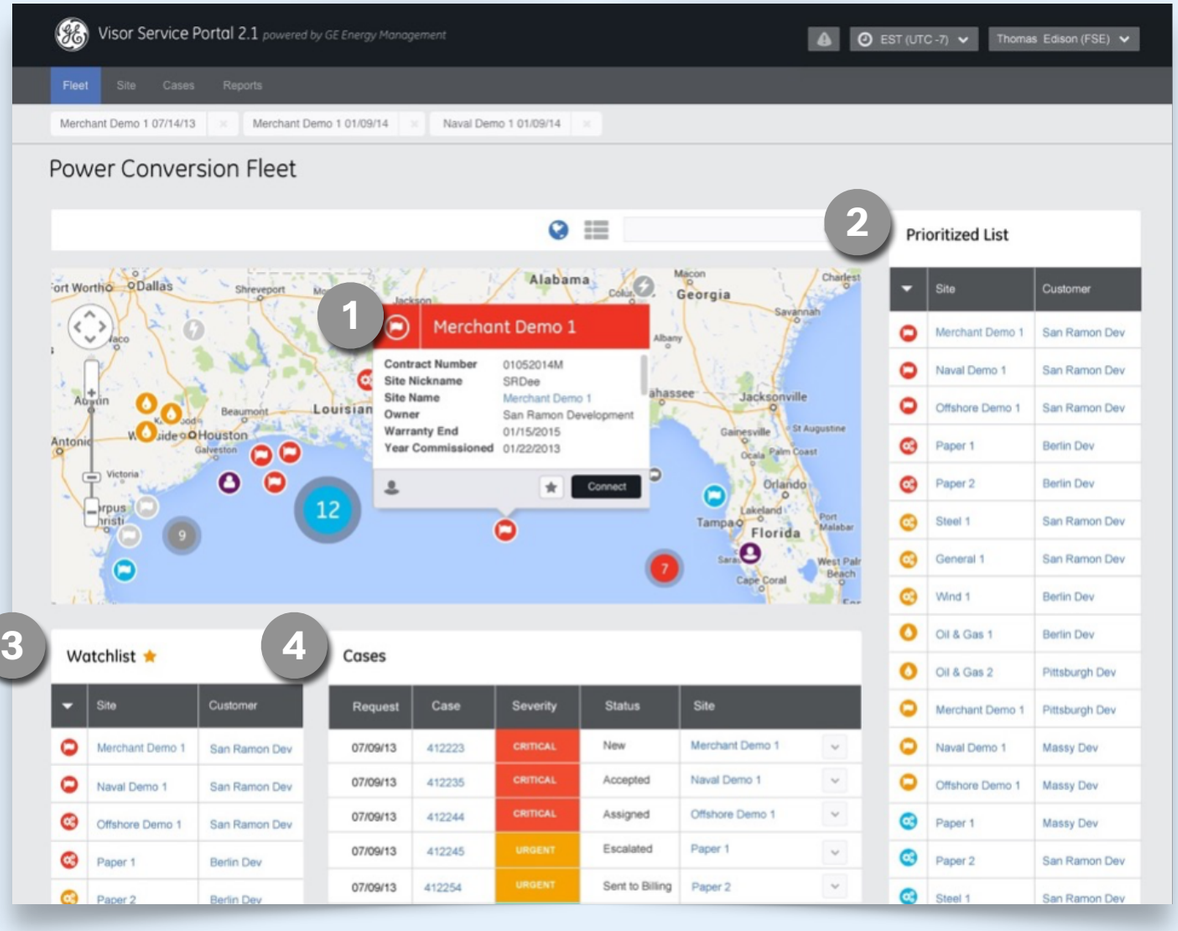
SOLUTION

Defined the product and service vision to align system architecture with the roadmap

Provide leaders with visuals for **fleet-wide situational awareness** of assets, FSE locations, and capabilities.

1. Display high-level asset business stats on rollover. Allow FSEs like Scott to **connect with a single click!**
2. Fleet Supervisors **prioritize assets by value to the success of the business.**
3. Personal “Watchlist” allows Scott to track the **assets he refers to most often.**
4. For Supervisors, a **list of open cases** to ensure the reduction of customer downtime.

Fleet Page Vision



VISION

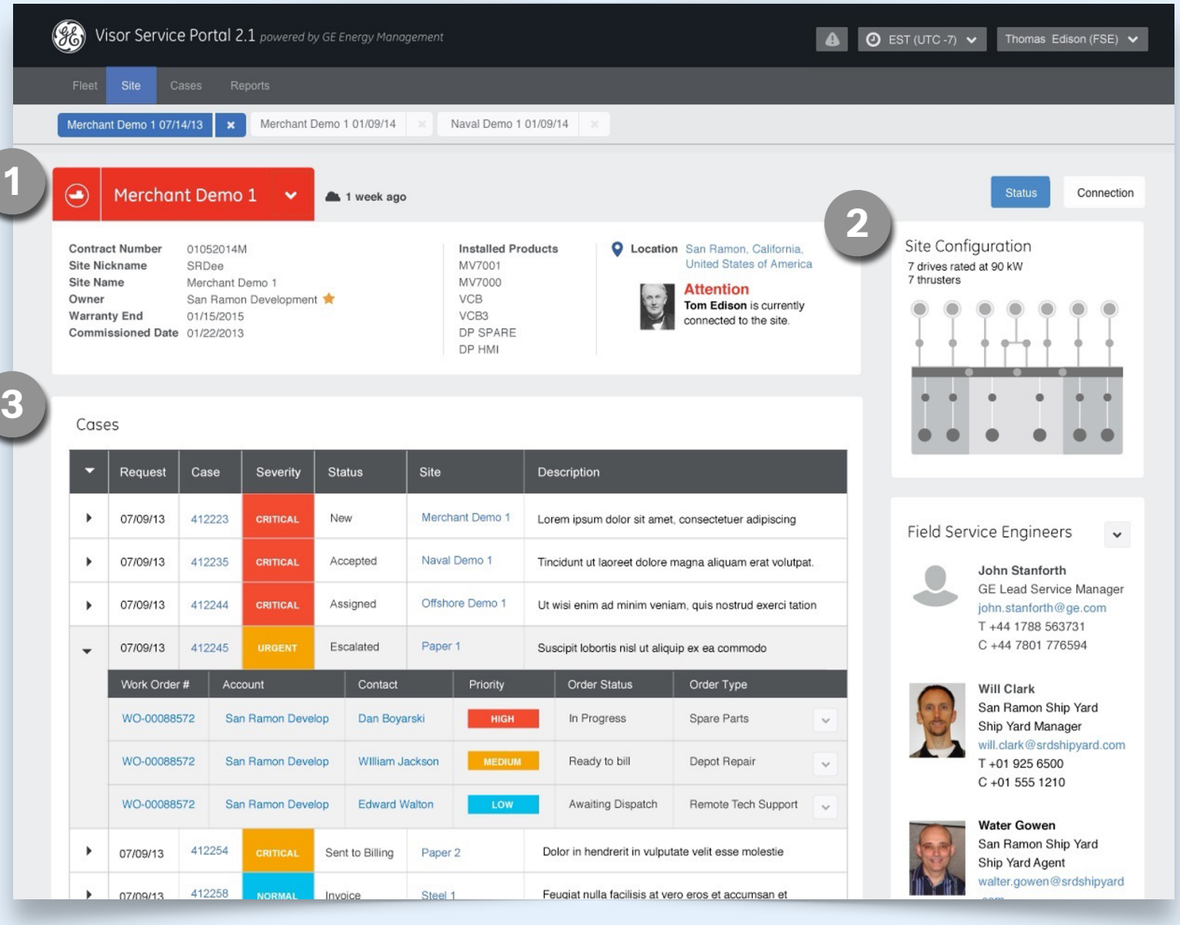
Asset drill-down reveals details once buried in binders and handwritten notes

1. Site block provides orientation and lists **which serviceable products are installed.**

2. A high level electrical site schematic orients the FSE to how the asset is configured.
   - How many thrusters are there?
   - Minimum generators to maintain power?
   - Which are down for maintenance?

3. Scott the FSE can see **what everyone on his team is currently working on** for this asset. (eg. If the customer has them on the phone and wants a status update.)

Asset Page Vision



VISION

North Star: A data historian visualization layering all diagnostic data sources

Assuming Scott has a stable connection or direct access to the Visor Connect Box, I envisioned a unified asset diagnostics tool with a **full time-series viewer.**

FSEs could jump to any timestamp and review data without cross-referencing multiple tools and sources.

1. Status of all power conversion systems and alarms at the time stamp of the position of the “play head.”
2. Complete **data historian signal processing tool,** including historical values and conditions … with tag overlays and groupings.
3. Faceted search / sort / filterable list of alarms, events, logs, and notes.

Data Historian Tools Vision



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User Research and Rapid Prototyping provided insights which accelerated our ability to deliver valuable outcomes to customers

Key Outcomes & Results

- Prioritization of the trip report allowed the team to deliver **30 days ahead of schedule.**
- Won a **$180M, 10-year service contract. Doubled sales in 2 years**
- Access to the Trip Report **reduced from 8 hours to 15 minutes!**
- Because VSP now reduces asset downtime and **improves availability**, it helps to close deals.
- We helped **build the reputation of GE Digital** with the GE Power Conversion and GE Oil & Gas by **earning their trust** with quantifiable results.

UX Lessons

- **Including end users in the process** enabled us to pivot to their needs. Our responsiveness **eased adoption.**
- Engineering led process risks building features without value. There's **no better source of insights than the frontline workers.**
- **"Get out of the building"** really works! It's risky to design marine field software from a white board.

Impact beyond MVP

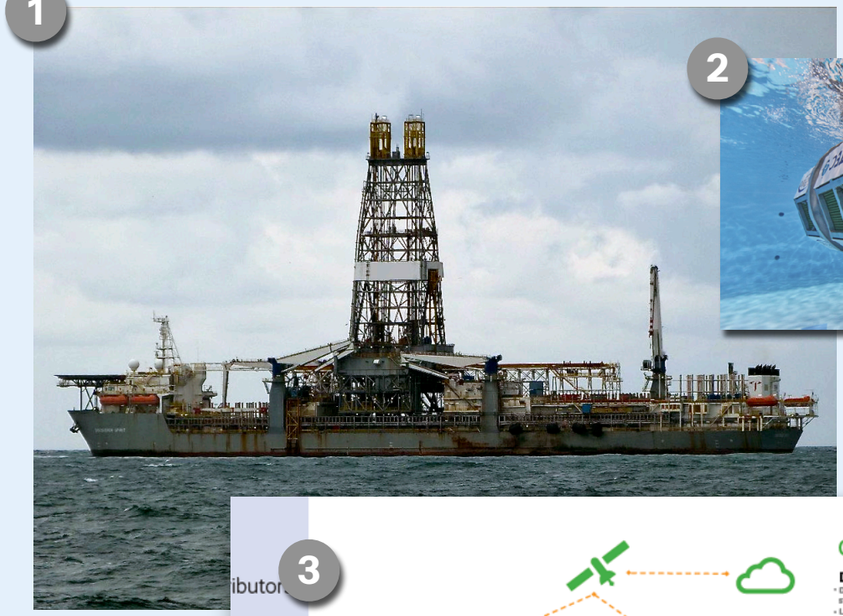
The success of my data historian work prompted a promotion to lead data visualization for the APM (Asset Performance Management) team.

1. Invited to **interview electrical engineers, bridge crew, and oil drillers** on a customer's drill ship.

2. **Helicopter Underwater Escape Training** required to visit a drill ship.

3. Our UX research insights defined the **GE Power Conversion Marine Product Roadmap.**

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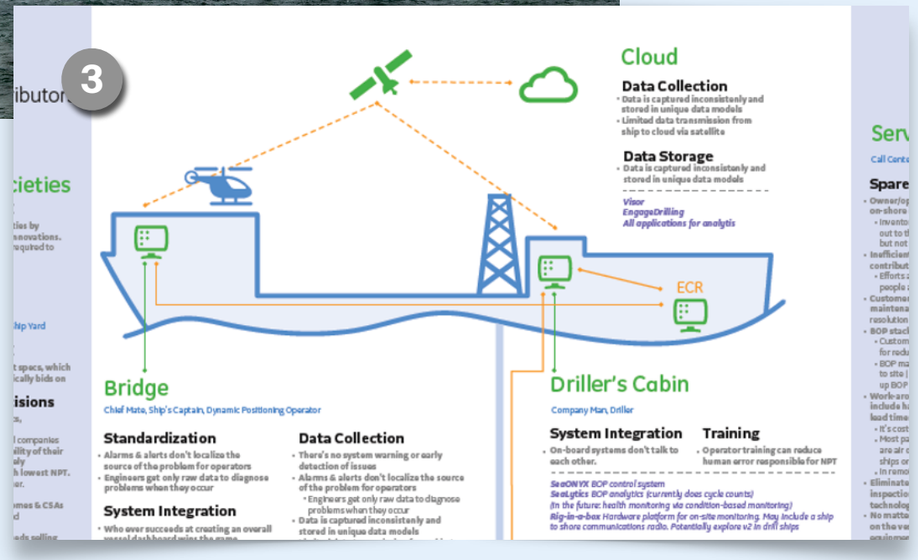


2



Stock photo

3



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THANKS FOR YOUR TIME!

Let’s discuss how I can provide AI augmented user-centered design to give the people who run your customers’ critical systems superpowers.



Contact Information:

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